KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE SCIENCES
DEPARTMENT OF MEDICAL ENGINEERING

FINAL QUALIFYING EXAMINATION
FOR
CERTIFICATE IN MEDICAL ENGINEERING TECHNOLOGY

PAPER: HOSPITAL PLANT & BUILDING SERVICES THEORY

DATE: 22nd February 2022 **TIME: 3 HOURS (9:00AM – 12:00 PM)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

HOSPITAL PLANT AND BUILDING SERVICES

SECTION 1 MULTIPLE CHOICE QUESTION 40 MARKS

1. Which of the following statement best describe shallow wells?
   a) Are wells where water is found
   b) Are wells where its water is found below the first impervious rock
   c) Are wells where its water is found above the first imperious strata
   d) Are wells where its water is tapped above the first pervious strata

2. The following are gas installation accessories which one is NOT?
   a) Pressure governors
   b) Pressure gauges
   c) Manifolds and multi-points
   d) Pressure release valve

3. What is the general safety requirement in laundry equipment?
   a) Emergency stop
   b) Locking mechanism
   c) Mounting
   d) All of the above

4. Which one below does not correctly describes the electrical installation safety requirement for washing machine?
   a) All metallic parts must be earthed
   b) Current operated ELCB must be connected
   c) Main switches should be placed at convenient height
   d) Gear, pulleys and v-belts are to be covered by guards

5. Which of the following is true of a positive displacement pump?
   a) Maintain a constant pressure
   b) Move a constant volume of fluid
   c) Vary the flow with Vpm
   d) Vary the flow with gear changes

6. Chillers, deep freezers and cold-rooms are used in the kitchen for the purposes of?
   a) Food processing
   b) Food preparation
   c) Food preservation
   d) Food processing and preservation

7. What causes acidity in water?
   a) Mineral acids
   b) Free carbon dioxide
   c) Iron sulphate
   d) All of the above

8. What is water-hammer?
   a) It is a way of minimizing water usage
   b) It is mechanical device
   c) It is a loud noise created when a rapid flow of water suddenly stopped
   d) All of the above
9. By boiling water, hardness can be removed if it is due to?
   a) Calcium sulphate
   b) Magnesium sulphate
   c) Calcium nitrate
   d) Calcium bicarbonate
10. Which one of the following is a tool used to make a copper joint?
   a) Swigging tool
   b) Surface plate
   c) Bending tool
   d) Flaring kit
11. What is the boiling point of ammonia?
   a) – 100°c
   b) -50°c
   c) -33.3°c
   d) -30.3°c
12. Which device is used to prevent unlit gas passing into combustion chamber of an appliance?
   a) Ignition device
   b) Flame failure device
   c) Solenoid valves
   d) Piezo igniter
13. In the four stroke engine the movement of the piston up and down in the cylinder is referred to as?
   a) Stroke
   b) Torque
   c) Cycle
   d) Induction
14. When an object is placed between the focal point and then lens, the image formed will -have which characteristic?
   a) Erect
   b) Diminished
   c) Real
   d) Formed at infinity
15. Which pump is preferred for small discharge at a high pressure?
   a) Centrifugal pump
   b) Axial flow pump
   c) Mixed flow pump
   d) Reciprocating pump

16. Which heat treatment process is done on plain carbon steel in order to soften it for machinability
   a) Carburizing
   b) Tempering
   c) Annealing
   d) Normalising
17. Freon group of refrigerants are?
   a) Inflammable
   b) Toxic
   c) Non-inflammable and toxic
   d) Non-toxic and non-inflammable

18. The opening and closing of inlet and exhaust valves of an in-controlled by which component of engine?
   a) Camshaft
   b) Connecting rods
   c) Valve spring
   d) Crank shaft
19. Hard water contains high concentration of which minerals?
   a) Iron and aluminum
   b) Chloride and sulphate
   c) Calcium and magnesium
   d) Chloride and aluminum

20. The colour code for carbon (iv) oxide five extinguishers is?
   a) Red
   b) Blue
   c) Black
   d) Brown
21. Impurities are removed from the boiler with the help of?
   a) Safely valve
   b) Blow off cock
   c) Stop valve
   d) Fusible plug

22. The refrigerant naming number for ammonia refrigerant is:
   a) R – 11
   b) R – 717
   c) R – 170
   d) R – 22
23. The type of engine used in the hospital standby generator is as
   a) External combination engine
   b) Paraffin engine
   c) Spark ignition engine
   d) Diesel engine

24. Kamau's refrigerator gives mild electrical shock when touched. What is the cause of this?
   a) Low current from the mains supply
   b) Higher current from the main supply
   c) Voltage fluctuation from the main supply
   d) Improper earthling of the mains supply
25. The air-fuel ratio of the petrol engine is controlled by?
   a) Fuel pump
   b) Governor
   c) Injector
   d) Carburetor

26. Stainless steel has the following properties except?
   a) It is resistant to rust
   b) It is good conductor of heat and electricity
   c) It is ductile
   d) It attacks magnet
27. Which of the following is safely requirement in laundry equipment?
   a) Door gasket
   b) Locking mechanism
   c) Butter rings
   d) Shipping nuts

28. What happens during suction stroke of a chisel engine?
   a) Only air is drawn into the combustion chamber
   b) Atomised diesel fuel is drawn into the combustion chamber
   c) A mixer of air and diesel fuel is drawn into the combustion chamber
   d) Removal of all the burnt gases from the cylinder to receive the fresh charge
29. Oven is a machine found in the kitchen and belongs to which equipment?
   a) Heating equipment
   b) Preservation equipment
   c) Preparation equipment
   d) None of the above

30. Which one of the following is not application of Bernoulli's principle?
   a) Working principles of a spry gun
   b) Working principle of liquid hydrometer
   c) Working principle of a Bunsen burner
   d) Working principle of carburettor
31. A device used to put off fire in the furnace of the boiler when the level of water in boiler falls to unsafe limit is?
   a) Blow off cork
   b) Fusible plug
   c) Super heater
   d) Economizer

32. A system of forces which when all the forces are extended pass through a single tended pass through a single point is called?
   a) Concurrent force system
   b) Coplanar force system
   c) Collinear force system
   d) Couple force system
33. Which one of the following is reason for installation of stop valve in steam boilers?
   a) Used to regulate flow of steam from boiler to steam pipes
   b) Used to shut off the steam completely when required
   c) It is fitted easy maintenance of system
   d) All of the above

34. What is the purpose of mechanical gate valve in cold water system?
   a) Control the flow of water
   b) Control the water level
   c) Connected at the draw off point
   d) Allow water to flow in one direction
35. Which one the following is a reason for mixing oil into the gas in a two-stroke engine?
   a) The holes in the cylinder was prevent the use of rings to seal the combustion chamber
   b) Holes in the lower part of cylinder wall lets in gas and air
   c) Holes in cylinder wall lest in oil
   d) A two-stroke engine has no moving valves

36. Which one of the following components is not found in a hydraulic system?
   a) Pump
   b) Actuator
   c) Compressor
   d) Motor
37. Steel pipes used for steam installation should be
   a) Galvanized and heavy
   b) Light and grey
   c) Black and heavy
   d) Black and light

38. Which hospital equipment makes use of gear pinions and levers in their operations?
   a) Operation theatre table
   b) Auto clave
   c) Dryers
   d) Potato peelers
39. Which of the following statements best describes the characteristic of non-aerated burners?
   a) The gas passes through an ejector nozzle and glass air with it
   b) The air-gas mixture in ignited of the burner head
   c) The fuel air mixture in ignited of the burner head for combustion only
   d) The gas does not mix with the air on its way to the burner head

40. What is the PH of pure water?
   a) 6.0
   b) 14.0
   c) 7.0
   d) 1.0

SECTION 2 SHORT ANSWER QUESTIONS 40 MARKS

1	a	State three types of flame failure devices used in gas burners	3 marks
	b	List two properties of liquidified petroleum gas	2 marks
2	a	State the sequence of starting the boiler	3 marks
	b	List two factors that affect the selection of a boiler	2 marks
3	a	State four advantages of hydro extractor machine in laundry department	4 marks
	b	State the purpose of hydro-extractor machine	1 mark
4		State five application of refrigeration in the Hospital	5 marks
5	a	State function of the following equipment refrigeration	
	i	System analyser	1 mark
	ii	Line tap valve	1 mark
	b	Name the function of the following components used in the refrigeration	
	i	T.E.V	1 mark
	ii	Capillary tube	1 mark
	iii	Filter	1 mark
6		State five advantages of a two stroke engine over a four stroke engine	5 marks
7		Outline the sequence of operation of petrol engine	5 marks
8		Explain the function of the following workshop tools	5 marks
	i	Anvil	
	ii	Bench vice	
	iii	Ball peen hammer	
	v	Water pump pliers	
	v	Pipe-wrench	

SECTION 3 LONG ANSWER QUESTION 20 MARKS

1	a	Explain five differences between window air conditioner and split unit air conditioner	10 marks
	b	With the aid of diagram describe how centrifugal pump works	6 marks
	c	Explain the procedure of adding refrigerant in the refrigeration system	4 marks